

5.4 SCALAR POINT FUNCTION

If for each point P of a region R , there corresponds a scalar denoted by $f(P)$, then f is called a “scalar point function” for the region R .

Example 1. The temperature $f(P)$ at any point P of a certain body occupying a certain region R is a scalar point function.

Example 2. The distance of any point $P(x, y, z)$ in space from a fixed point (x_0, y_0, z_0) is a scalar function.

$$f(P) = \sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}.$$

Scalar field

(U.P.T.U., 2001)

Scalar field is a region in space such that for every point P in this region, the scalar function f associates a scalar $f(P)$.

5.5 VECTOR POINT FUNCTION

If for each point P of a region R , there corresponds a vector $\vec{f}(P)$ then $\vec{f}$ is called “vector point function” for the region R .

Example. If the velocity of a particle at a point P , at any time t be $\vec{f}(P)$, then $\vec{f}$ is a vector point function for the region occupied by the particle at time t .

If the coordinates of P be (x, y, z) then

$$\vec{f}(P) = f_1(x, y, z) \mathbf{i} + f_2(x, y, z) \mathbf{j} + f_3(x, y, z) \mathbf{k}.$$

Vector field

(U.P.T.U., 2001)

Vector field is a region in space such that with every point P in the region, the vector function $\vec{f}$ associates a vector $\vec{f}(P)$.

Del operator: The linear vector differential (Hamiltonian) operator “del” defined and

denoted as

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

This operator also read nabla. It is not a vector but combines both differential and vectorial properties analogous to those of ordinary vectors.

5.6 GRADIENT OR SLOPE OF SCALAR POINT FUNCTION

If $f(x, y, z)$ be a scalar point function and continuously differentiable then the vector

$$\vec{\nabla} f = \mathbf{i} \frac{\partial f}{\partial x} + \mathbf{j} \frac{\partial f}{\partial y} + \mathbf{k} \frac{\partial f}{\partial z}$$

is called the gradient of f and is written as $\text{grad } f$.

(U.P.T.U., 2006)

Thus

$$\text{grad } f = \mathbf{i} \frac{\partial f}{\partial x} + \mathbf{j} \frac{\partial f}{\partial y} + \mathbf{k} \frac{\partial f}{\partial z} = \vec{\nabla} f$$

It should be noted that $\vec{\nabla} f$ is a vector whose three components are $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, $\frac{\partial f}{\partial z}$. Thus, if f is a scalar point function, then $\vec{\nabla} f$ is a vector point function.

5.7 GEOMETRICAL MEANING OF GRADIENT, NORMAL

Consider any point P in a region through which a scalar field $f(x, y, z) = c$ defined. Suppose that $\vec{\nabla}f \neq 0$ at P and that there is a $f = \text{const.}$ surface S through P and a tangent plane T ; for instance, if f is a temperature field, then S is an isothermal surface (level surface). If $\hat{n}$, at P , is chosen as any vector in the tangent plane T , then surely $\frac{df}{dS}$ must be zero.

$$\text{Since } \frac{df}{dS} = \vec{\nabla}f \cdot \hat{n} = 0$$

for every $\hat{n}$ at P in the tangent plane, and both $\vec{\nabla}f$ and $\hat{n}$ are non-zero, it follows that $\vec{\nabla}f$ is normal to the tangent plane T and hence to the surface S at P .

If letting $\hat{n}$ be in the tangent plane, we learn that $\vec{\nabla}f$ is normal to S , then to seek additional information about $\vec{\nabla}f$ it seems logical to let $\hat{n}$ be along the normal line at P .

$$\text{Then } \frac{df}{ds} = \frac{df}{dN}, \hat{N} = \hat{n} \text{ then}$$

$$\frac{df}{dN} = \vec{\nabla}f \cdot \hat{N} = |\vec{\nabla}f| \cdot 1 \cos 0 = |\vec{\nabla}f|.$$

So that the magnitude of $\vec{\nabla}f$ is the directional derivative of f along the normal line to S , in the direction of increasing f .

Hence, "The gradient $(\vec{\nabla}f)$ of scalar field $f(x, y, z)$ at P is vector normal to the surface $f = \text{const.}$ and has a magnitude is equal to the directional derivative $\frac{df}{dN}$ in that direction.

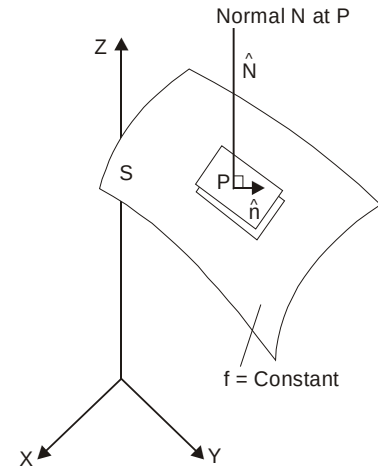


Fig. 5.2

5.8 DIRECTIONAL DERIVATIVE

Let $f = f(x, y, z)$ then the partial derivatives $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$ are the derivatives (rates of change) of f in the direction of the coordinate axes OX, OY, OZ respectively. This concept can be extended to define a derivative of f in a "given" direction $\overrightarrow{PQ}$.

Let P be a point in space and $\hat{b}$ be a unit vector from P in the given direction. Let s be the arc length measured from P to another point Q along the ray C in the direction of $\hat{b}$. Now consider

$$f(s) = f(x, y, z) = f\{x(s), y(s), z(s)\}$$

$$\text{Then } \frac{df}{ds} = \frac{\partial f}{\partial x} \frac{dx}{ds} + \frac{\partial f}{\partial y} \frac{dy}{ds} + \frac{\partial f}{\partial z} \frac{dz}{ds} \quad \dots(i)$$

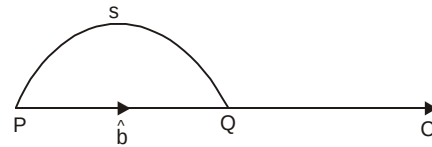


Fig. 5.3

Here $\frac{df}{ds}$ is called directional derivative of f at P in the direction $\hat{b}$ which gives the rate of change of f in the direction of b .

$$\text{Since, } \frac{dx}{ds}\hat{i} + \frac{dy}{ds}\hat{j} + \frac{dz}{ds}\hat{k} = \hat{b} = \text{unit vector} \quad \dots(ii)$$

Eqn. (i) can be rewritten as

$$\begin{aligned} \frac{df}{ds} &= \left(\hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} \right) \cdot \left(\frac{dx}{ds}\hat{i} + \frac{dy}{ds}\hat{j} + \frac{dz}{ds}\hat{k} \right) \\ \frac{df}{ds} &= \left[\left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) f \right] \cdot \hat{b} = \nabla f \cdot \hat{b} \quad \dots(iii) \end{aligned}$$

Thus the directional derivative of f at P is the component (dot product) of ∇f in the direction of (with) unit vector $\hat{b}$.

Hence the directional derivative in the direction of any unit vector $\vec{a}$ is

$$\boxed{\frac{df}{ds} = \nabla f \cdot \left(\frac{\vec{a}}{|\vec{a}|} \right)}$$

Normal derivative $\frac{df}{dn} = \nabla f \cdot \hat{n}$, where $\hat{n}$ is the unit normal to the surface $f = \text{constant}$.

5.9 PROPERTIES OF GRADIENT

Property I: $(\vec{a} \cdot \nabla)f = \vec{a} \cdot (\nabla f)$

Proof: L.H.S. = $(\vec{a} \cdot \nabla)f$

$$\begin{aligned} &= \left\{ \vec{a} \cdot \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \right\} f \\ &= \left\{ (\vec{a} \cdot \hat{i}) \frac{\partial}{\partial x} + (\vec{a} \cdot \hat{j}) \frac{\partial}{\partial y} + (\vec{a} \cdot \hat{k}) \frac{\partial}{\partial z} \right\} f \\ &= (\vec{a} \cdot \hat{i}) \frac{\partial f}{\partial x} + (\vec{a} \cdot \hat{j}) \frac{\partial f}{\partial y} + (\vec{a} \cdot \hat{k}) \frac{\partial f}{\partial z} \quad \dots(i) \end{aligned}$$

R.H.S. = $\vec{a} \cdot (\nabla f)$

$$\begin{aligned} &= \vec{a} \cdot \left(\hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} \right) \\ &= (a \cdot i) \frac{\partial f}{\partial x} + (a \cdot j) \frac{\partial f}{\partial y} + (a \cdot k) \frac{\partial f}{\partial z} \quad \dots(ii) \end{aligned}$$

From (i) and (ii),

$$\boxed{(\vec{a} \cdot \nabla)f = \vec{a} \cdot (\nabla f)}$$

Property II: Gradient of a Constant

The necessary and sufficient condition that scalar point function ϕ is a constant is that $\nabla\phi = 0$.

Proof: Let $\phi(x, y, z) = c$

Then,
$$\frac{\partial\phi}{\partial x} = \frac{\partial\phi}{\partial y} = \frac{\partial\phi}{\partial z} = 0$$

$$\begin{aligned}\therefore \nabla\phi &= i\frac{\partial\phi}{\partial x} + j\frac{\partial\phi}{\partial y} + k\frac{\partial\phi}{\partial z} \\ &= i.0 + j.0 + k.0 \\ &= 0.\end{aligned}$$

Hence, the condition is necessary.

Conversely: Let $\nabla\phi = 0$.

Then,
$$i\frac{\partial\phi}{\partial x} + j\frac{\partial\phi}{\partial y} + k\frac{\partial\phi}{\partial z} = 0.i + 0.j + 0.k.$$

Equating the coefficients of i, j, k .

On both sides, we get $\frac{\partial\phi}{\partial x} = 0, \frac{\partial\phi}{\partial y} = 0, \frac{\partial\phi}{\partial z} = 0$

$\Rightarrow \phi$ is independent of x, y, z

$\Rightarrow \phi$ is constant.

Hence, the condition is sufficient.

Property III: Gradient of the Sum of Difference of Two Functions

If f and g are any two scalar point functions, then

$$\begin{aligned}\nabla(f \pm g) &= \nabla f \pm \nabla g \\ \text{or } \text{grad}(f \pm g) &= \text{grad } f \pm \text{grad } g\end{aligned}$$

Proof:
$$\begin{aligned}\nabla(f \pm g) &= \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)(f \pm g) \\ &= i\frac{\partial}{\partial x}(f \pm g) + j\frac{\partial}{\partial y}(f \pm g) + k\frac{\partial}{\partial z}(f \pm g) \\ &= i\left(\frac{\partial f}{\partial x} \pm \frac{\partial g}{\partial x}\right) + j\left(\frac{\partial f}{\partial y} \pm \frac{\partial g}{\partial y}\right) + k\left(\frac{\partial f}{\partial z} \pm \frac{\partial g}{\partial z}\right) \\ &= \left(i\frac{\partial f}{\partial x} + j\frac{\partial g}{\partial y} + k\frac{\partial f}{\partial z}\right) \pm \left(i\frac{\partial g}{\partial x} + j\frac{\partial g}{\partial y} + k\frac{\partial g}{\partial z}\right)\end{aligned}$$

$$\nabla(f \pm g) = \text{grad } f \pm \text{grad } g.$$

Property IV: Gradient of the Product of Two Functions

If f and g are two scalar point functions, then

$$\begin{aligned}\nabla(fg) &= f\nabla g + g\nabla f \\ \text{or } \text{grad}(fg) &= f(\text{grad } g) + g(\text{grad } f).\end{aligned}$$

Proof:
$$\nabla(fg) = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)(fg)$$

$$\begin{aligned}
&= i \frac{\partial}{\partial x} (fg) + j \frac{\partial}{\partial y} (fg) + k \frac{\partial}{\partial z} (fg) \\
&= i \left(f \frac{\partial g}{\partial x} + g \frac{\partial f}{\partial x} \right) + j \left(f \frac{\partial g}{\partial y} + g \frac{\partial f}{\partial y} \right) + k \left(f \frac{\partial g}{\partial z} + g \frac{\partial f}{\partial z} \right) \\
&= f \left(i \frac{\partial g}{\partial x} + j \frac{\partial g}{\partial y} + k \frac{\partial g}{\partial z} \right) + g \left(i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z} \right)
\end{aligned}$$

$$\boxed{\nabla(fg) = f\nabla g + g\nabla f.}$$

Property V: Gradient of the Quotient of Two Functions

If f and g are two scalar point functions, then

$$\nabla \left(\frac{f}{g} \right) = \frac{g\nabla f - f\nabla g}{g^2}, \quad g \neq 0$$

or

$$\text{grad} \left(\frac{f}{g} \right) = \frac{g(\text{grad } f) - f(\text{grad } g)}{g^2}, \quad g \neq 0.$$

Proof:

$$\begin{aligned}
\nabla \left(\frac{f}{g} \right) &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \left(\frac{f}{g} \right) \\
&= i \frac{\partial}{\partial x} \left(\frac{f}{g} \right) + j \frac{\partial}{\partial y} \left(\frac{f}{g} \right) + k \frac{\partial}{\partial z} \left(\frac{f}{g} \right) \\
&= i \frac{g \frac{\partial f}{\partial x} - f \frac{\partial g}{\partial x}}{g^2} + j \frac{g \frac{\partial f}{\partial y} - f \frac{\partial g}{\partial y}}{g^2} + k \frac{g \frac{\partial f}{\partial z} - f \frac{\partial g}{\partial z}}{g^2} \\
&= \frac{g \left[i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z} \right] - f \left[i \frac{\partial g}{\partial x} + j \frac{\partial g}{\partial y} + k \frac{\partial g}{\partial z} \right]}{g^2}
\end{aligned}$$

$$\boxed{\nabla \left(\frac{f}{g} \right) = \frac{g\nabla f - f\nabla g}{g^2}.}$$

Example 1. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then show that

(U.P.T.U., 2007)

(i) $\nabla(\vec{a} \cdot \vec{r}) = \vec{a}$, where $\vec{a}$ is a constant vector

(ii) $\text{grad } r = \frac{\vec{r}}{r}$

(iii) $\text{grad} \frac{1}{r} = -\frac{\vec{r}}{r^3}$

(iv) $\text{grad } r^n = nr^{n-2} \vec{r}$, where $r = |\vec{r}|$.

Sol. (i) Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$,

then

$$\vec{a} \cdot \vec{r} = (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \cdot (x\hat{i} + y\hat{j} + z\hat{k}) = a_1x + a_2y + a_3z.$$

$$\begin{aligned}\therefore \nabla(\vec{a} \cdot \vec{r}) &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (a_1 x + a_2 y + a_3 z) \\ &= a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k} = \vec{a}. \quad \text{Hence proved.}\end{aligned}$$

$$\begin{aligned}(ii) \quad \text{grad } r = \Delta r &= \Sigma \hat{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{1/2} \\ &= \Sigma \hat{i} \frac{x}{(x^2 + y^2 + z^2)^{1/2}} = \Sigma \hat{i} \frac{x}{r} = r\end{aligned}$$

Hence, $\text{grad } r = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{r} = \frac{\vec{r}}{r} = \hat{r}.$

$$\begin{aligned}(iii) \quad \text{grad} \left(\frac{1}{r} \right) &= \nabla \left(\frac{1}{r} \right) = \frac{\partial}{\partial r} \left(\frac{1}{r} \right) \vec{r} = \frac{-1}{r^2} \vec{r} \\ &= -\frac{\vec{r}}{r^3}. \quad \text{Proved.}\end{aligned}$$

(iv) Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}.$

$$\begin{aligned}\text{Now,} \quad \text{grad } r^n &= \nabla r^n = \Sigma \hat{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{n/2} \\ &= n (x^2 + y^2 + z^2)^{n/2-1} (x\hat{i} + y\hat{j} + z\hat{k}) \\ &= n (x^2 + y^2 + z^2)^{(n-1)/2} \frac{(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{1/2}} \\ &= nr^{n-1} \frac{\vec{r}}{r} \\ &= nr^{n-2} \vec{r}.\end{aligned}$$

Example 2. If $f = 3x^2y - y^3z^2$, find $\text{grad } f$ at the point $(1, -2, -1)$. (U.P.T.U., 2006)

$$\begin{aligned}\text{Sol.} \quad \text{grad } f &= \vec{\nabla} f = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (3x^2y - y^3z^2) \\ &= i \frac{\partial}{\partial x} (3x^2y - y^3z^2) + j \frac{\partial}{\partial y} (3x^2y - y^3z^2) + k \frac{\partial}{\partial z} (3x^2y - y^3z^2) \\ &= i(6xy) + j(3x^2 - 3y^2z^2) + k(-2y^3z) \\ \text{grad } \phi \text{ at } (1, -2, -1) &= i(6)(1)(-2) + j[(3)(1) - 3(4)(1)] + k(-2)(-8)(-1) \\ &= -12i - 9j - 16k.\end{aligned}$$

Example 3. Find the directional derivative of $\frac{1}{r}$ in the direction $\vec{r}$ where $\vec{r} = xi + yj + zk$. (U.P.T.U., 2002, 2005)

$$\text{Sol. Here } f(x, y, z) = \frac{1}{r} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} = (x^2 + y^2 + z^2)^{-1/2}$$

$$\text{Now } \vec{\nabla} f = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (x^2 + y^2 + z^2)^{-1/2}$$

$$\begin{aligned}
&= \frac{\partial}{\partial x}(x^2 + y^2 + z^2)^{-1/2} i + \frac{\partial}{\partial y}(x^2 + y^2 + z^2)^{-1/2} j + \frac{\partial}{\partial z}(x^2 + y^2 + z^2)^{-1/2} k \\
&= \left\{ -\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2} 2x \right\} i + \left\{ -\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2} 2y \right\} j + \left\{ -\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2} 2z \right\} k \\
&= \frac{-(xi + yj + zk)}{(x^2 + y^2 + z^2)^{3/2}}
\end{aligned}$$

and $\hat{a}$ = unit vector in the direction of $xi + yj + zk$

$$= \frac{xi + yj + zk}{\sqrt{x^2 + y^2 + z^2}}$$

$$\left| \text{As } \hat{a} = \frac{\vec{r}}{|\vec{r}|} \right|$$

$$\begin{aligned}
\therefore \text{ Directional derivative} = \nabla f \cdot \hat{a} &= -\frac{xi + yj + zk}{(x^2 + y^2 + z^2)^{3/2}} \cdot \frac{xi + yj + zk}{(x^2 + y^2 + z^2)^{1/2}} \\
&= -\frac{(xi + yj + zk)^2}{(x^2 + y^2 + z^2)^2} = -\left(\frac{xi + yj + zk}{x^2 + y^2 + z^2} \right)^2.
\end{aligned}$$

Example 4. Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2i - j - 2k$. In what direction the directional derivative will be maximum and what is its magnitude? Also find a unit normal to the surface $x^2yz + 4xz^2 = 6$ at the point $(1, -2, -1)$.

Sol. $\phi = x^2yz + 4xz^2$

$$\therefore \frac{\partial \phi}{\partial x} = 2xyz + 4z^2$$

$$\frac{\partial \phi}{\partial y} = x^2z,$$

$$\frac{\partial \phi}{\partial z} = x^2y + 8xz$$

$$\begin{aligned}
\text{grad } \phi &= i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z} \\
&= (2xyz + 4z^2) i + (x^2z) j + (x^2y + 8xz) k \\
&= 8i - j - 10k \text{ at the point } (1, -2, -1)
\end{aligned}$$

Let $\hat{a}$ be the unit vector in the given direction.

Then
$$\hat{a} = \frac{2i - j - 2k}{\sqrt{4 + 1 + 4}} = \frac{2i - j - 2k}{3}$$

$$\begin{aligned}
\therefore \text{ Directional derivative } \frac{d\phi}{ds} &= \nabla \phi \cdot \hat{a} \\
&= (8i - j - 10k) \cdot \left(\frac{2i - j - 2k}{3} \right) \\
&= \frac{16 + 1 + 20}{3} = \frac{37}{3}.
\end{aligned}$$

Again, we know that the directional derivative is maximum in the direction of normal which is the direction of $\text{grad } \phi$. Hence, the directional derivative is maximum along $\text{grad } \phi = 8i - j - 10k$.

Further, maximum value of the directional derivative

$$\begin{aligned} &= |\text{grad } \phi| \\ &= |8i - j - 10k| \\ &= \sqrt{64+1+100} = \sqrt{165}. \end{aligned}$$

Again, a unit vector normal to the surface

$$\begin{aligned} &= \frac{\text{grad } \phi}{|\text{grad } \phi|} \\ &= \frac{8i - j - 10k}{\sqrt{165}}. \end{aligned}$$

Example 5. What is the greatest rate of increase of $u = xyz^2$ at the point $(1, 0, 3)$?

Sol.

$$u = xyz^2$$

$$\begin{aligned} \therefore \text{grad } u &= i \frac{\partial u}{\partial x} + j \frac{\partial u}{\partial y} + k \frac{\partial u}{\partial z} \\ &= yz^2 i + xz^2 j + 2xyz k \\ &= 9j \text{ at } (1, 0, 3) \text{ point.} \end{aligned}$$

Hence, the greatest rate of increase of u at $(1, 0, 3)$

$$\begin{aligned} &= |\text{grad } u| \text{ at } (1, 0, 3) \text{ point.} \\ &= |9j| = 9. \end{aligned}$$

Example 6. Find the directional derivative of

$$\phi = (x^2 + y^2 + z^2)^{-1/2}$$

at the points $(3, 1, 2)$ in the direction of the vector $yz i + zx j + xy k$.

Sol.

$$\phi = (x^2 + y^2 + z^2)^{-1/2}$$

$$\begin{aligned} \therefore \text{grad } \phi &= i \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-1/2} + j \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{-1/2} \\ &\quad + k \frac{\partial}{\partial z} (x^2 + y^2 + z^2)^{-1/2} \\ &= i \left[-\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (2x) \right] + j \left[-\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (2y) \right] \\ &\quad + k \left[-\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (2z) \right] \\ &= -\frac{xi + yj + zk}{(x^2 + y^2 + z^2)^{3/2}} \\ &= -\frac{3i + j + 2k}{(9 + 1 + 4)^{3/2}} \text{ at } (3, 1, 2) \\ &= -\frac{3i + j + 2k}{14\sqrt{14}} \text{ at } (3, 1, 2) \end{aligned}$$

Let $\hat{a}$ be the unit vector in the given direction, then

$$\begin{aligned}\hat{a} &= \frac{yzi + zxj + xyk}{\sqrt{y^2z^2 + z^2x^2 + x^2y^2}} \\ &= \frac{2i + 6j + 3k}{7} \text{ at } (3, 1, 2)\end{aligned}$$

Now,

$$\begin{aligned}\frac{d\phi}{ds} &= \hat{a} \cdot \text{grad } \phi \\ &= \left(\frac{2i + 6j + 3k}{7} \right) \cdot \left(-\frac{3i + j + 2k}{14\sqrt{14}} \right) \\ &= -\frac{(2)(3) + (6)(1) + (3)(2)}{7 \cdot 14\sqrt{14}} \\ &= -\frac{18}{7 \cdot 14\sqrt{14}} = -\frac{9}{49\sqrt{14}}.\end{aligned}$$

Example 7. Find the directional derivative of the function $\phi = x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of the line PQ , where Q is the point $(5, 0, 4)$.

Sol. Here

$$\text{Position vector of } P = i + 2j + 3k$$

$$\text{Position vector of } Q = 5i + 0j + 4k$$

$$\begin{aligned}\therefore \quad \overrightarrow{PQ} &= \text{Position vector of } Q - \text{Position vector of } P \\ &= (5i + 0j + 4k) - (i + 2j + 3k) \\ &= 4i - 2j + k.\end{aligned}$$

Let $\hat{a}$ be the unit vector along PQ , then

$$\hat{a} = \frac{4i - 2j + k}{\sqrt{16 + 4 + 1}} = \frac{4i - 2j + k}{\sqrt{21}}$$

Also,

$$\begin{aligned}\text{grad } \phi &= i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z} \\ &= 2x i - 2y j + 4z k \\ &= 2i - 4j + 12k \text{ at } (1, 2, 3)\end{aligned}$$

Hence,

$$\begin{aligned}\frac{d\phi}{ds} &= \hat{a} \cdot \text{grad } \phi \\ &= \left(\frac{4i - 2j + k}{\sqrt{21}} \right) \cdot (2i - 4j + 12k) \\ &= \frac{(4)(2) + (-2)(-4) + (1)(12)}{\sqrt{21}} \\ &= \frac{28}{\sqrt{21}}.\end{aligned}$$

Example 8. For the function

$$\phi = \frac{y}{x^2 + y^2},$$

find the magnitude of the directional derivative making an angle 30° with the positive X-axis at the point $(0, 1)$.

Sol. Here $\phi = \frac{y}{x^2 + y^2}$

$$\therefore \frac{\partial \phi}{\partial x} = -\frac{2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial \phi}{\partial y} = \frac{x^2 - y^2}{(x^2 + y^2)^2} \text{ and } \frac{\partial \phi}{\partial z} = 0$$

$$\therefore \text{grad } \phi = i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z}$$

$$= i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} \quad \left[\because \frac{\partial \phi}{\partial z} = 0 \right]$$

$$= \frac{-2xy}{(x^2 + y^2)^2} i + \frac{x^2 - y^2}{(x^2 + y^2)^2} j$$

$$= -j \text{ at } (0, 1).$$

Let $\hat{a}$ be the unit vector along the line making an angle 30° with the positive X-axis at the point $(0, 1)$, then

$$\hat{a} = \cos 30^\circ i + \sin 30^\circ j.$$

Hence, the directional derivative is given by

$$\frac{d\phi}{ds} = \hat{a} \cdot \text{grad } \phi$$

$$= (\cos 30^\circ i + \sin 30^\circ j) \cdot (-j)$$

$$= -\sin 30^\circ = -\frac{1}{2}.$$

Example 9. Find the values of the constants a, b, c so that the directional derivative of $\phi = axy^2 + byz + cz^2x^3$ at $(1, 2, -1)$ has a maximum magnitude 64 in the direction parallel to Z-axis.

Sol. $\phi = axy^2 + byz + cz^2x^3$

$$\therefore \text{grad } \phi = \frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j + \frac{\partial \phi}{\partial z} k$$

$$= (ay^2 + 3cz^2x^2) i + (2axy + bz) j + (by + 2czx^3) k$$

$$= (4a + 3c) i + (4a - b) j + (2b - 2c) k. \text{ at } (1, 2, -1).$$

Now, we know that the directional derivative is maximum along the normal to the surface, *i.e.*, along $\text{grad } \phi$. But we are given that the directional derivative is maximum in the direction parallel to Z-axis, *i.e.*, parallel to the vector k .

Hence, the coefficients of i and j in $\text{grad } \phi$ should vanish and the coefficient of k should be positive. Thus

$$4a + 3c = 0 \quad \dots(i)$$

$$4a - b = 0 \quad \dots(ii)$$

and $2b - 2c > 0$

$$b > c \quad \dots(iii)$$

Then $\text{grad } \phi = 2(b - c) k.$

Also, maximum value of the directional derivative = $|\text{grad } \phi|$

$$\Rightarrow 64 = |2(b-c)k| = 2(b-c) \quad [\because b > c]$$

$$\Rightarrow b - c = 32. \quad \dots(iv)$$

Solving equations (i), (ii) and (iv), we obtain

$$a = 6, b = 24, c = -8.$$

Example 10. If the directional derivative of $\phi = ax^2y + by^2z + cz^2x$, at the point (1, 1, 1) has maximum magnitude 15 in the direction parallel to the line $\frac{x-1}{2} = \frac{y-3}{-2} = \frac{z}{1}$, find the values of a , b and c . (U.P.T.U., 2001)

Sol. Given

$$\phi = ax^2y + by^2z + cz^2x$$

$$\begin{aligned} \therefore \vec{\nabla}\phi &= i\left(\frac{\partial\phi}{\partial x}\right) + j\left(\frac{\partial\phi}{\partial y}\right) + k\left(\frac{\partial\phi}{\partial z}\right) \\ &= i(2axy + cz^2) + j(ax^2 + 2byz) + k(by^2 + 2czx) \end{aligned}$$

$$\vec{\nabla}\phi \text{ at the point } (1, 1, 1) = i(2a + c) + j(a + 2b) + k(b + 2c).$$

We know that the maximum value of the directional derivative is in the direction of $\vec{\nabla}f$ i.e., $|\nabla\phi| = 15 \Rightarrow (2a + c)^2 + (2b + a)^2 + (2c + b)^2 = (15)^2$

But, the directional derivative is given to be maximum parallel to the line.

$$\frac{x-1}{2} = \frac{y-3}{-2} = \frac{z}{1}$$

$$\Rightarrow \frac{2a+c}{2} = \frac{2b+a}{-2} = \frac{2c+b}{1} \quad (i)$$

$$\Rightarrow 2a + c = -2b - a \Rightarrow 3a + 2b + c = 0$$

$$\text{and } 2b + a = -2(2c + b)$$

$$\Rightarrow 2b + a = -4c - 2b \Rightarrow a + 4b + 4c = 0 \quad (ii)$$

Solving (i) and (ii), we get

$$\frac{a}{4} = \frac{b}{-11} = \frac{c}{10} = k \text{ (say)}$$

$$\Rightarrow a = 4k, b = -11k, \text{ and } c = 10k.$$

$$\text{Now, } (2a + c)^2 + (2b + a)^2 + (2c + b)^2 = (15)^2$$

$$\Rightarrow (8k + 10k)^2 + (-22k + 4k)^2 + (20k - 11k)^2 = (15)^2$$

$$\Rightarrow k = \pm \frac{5}{9}$$

$$\Rightarrow a = \pm \frac{20}{9}, b = \pm \frac{55}{9} \text{ and } c = \pm \frac{50}{9}.$$

Example 11. Prove that $\nabla \int f(u) du = f(u) \nabla u$.

Sol. Let $\int f(u) du = F(u)$, a function of u so that $\frac{\partial F}{\partial u} = f(u)$

$$\text{Then } \nabla \int f(u) du = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) F$$

$$\begin{aligned}
&= i \frac{\partial F}{\partial x} + j \frac{\partial F}{\partial y} + k \frac{\partial F}{\partial z} \\
&= i \frac{\partial F}{\partial u} \frac{\partial u}{\partial x} + j \frac{\partial F}{\partial u} \frac{\partial u}{\partial y} + k \frac{\partial F}{\partial u} \frac{\partial u}{\partial z} \\
&= \frac{\partial F}{\partial u} \left(i \frac{\partial u}{\partial x} + j \frac{\partial u}{\partial y} + k \frac{\partial u}{\partial z} \right) \\
&= f(u) \nabla u. \quad \text{Hence proved.}
\end{aligned}$$

Example 12. Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$ (U.P.T.U., 2002)

Sol. Let

$$\phi_1 = x^2 + y^2 + z^2 - 9$$

$$\phi_2 = x^2 + y^2 - z - 3$$

$$\therefore \quad \vec{\nabla}\phi_1 = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (x^2 + y^2 + z^2 - 9) = 2xi + 2yj + 2zk$$

$$\nabla\phi_1 \text{ at the point } (2, -1, 2) = 4i - 2j + 4k \quad (i)$$

$$\vec{\nabla}\phi_2 = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (x^2 + y^2 - z - 3) = 2xi + 2yj - k$$

$$\nabla\phi_2 \text{ at the point } (2, -1, 2) = 4i - 2j - k \quad (ii)$$

Let θ be the angle between normals (i) and (ii)

$$(4i - 2j + 4k) \cdot (4i - 2j - k) = \sqrt{16+4+16} \sqrt{16+4+1} \cos \theta$$

$$16 + 4 - 4 = 6\sqrt{21} \cos \theta \Rightarrow 16 = 6\sqrt{21} \cos \theta$$

$$\Rightarrow \quad \cos \theta = \frac{8}{3\sqrt{21}} \Rightarrow \theta = \cos^{-1} \frac{8}{3\sqrt{21}}.$$

Example 13. If $u = x + y + z$, $v = x^2 + y^2 + z^2$, $w = yz + zx + xy$, prove that $\text{grad } u$, $\text{grad } v$ and $\text{grad } w$ are coplanar vectors. (U.P.T.U., 2002)

Sol.

$$\text{grad } u = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x + y + z) = \hat{i} + \hat{j} + \hat{k}$$

$$\text{grad } v = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x^2 + y^2 + z^2) = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

$$\begin{aligned}
\text{grad } w &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (yz + zx + xy) \\
&= \hat{i}(z+y) + \hat{j}(z+x) + \hat{k}(y+x)
\end{aligned}$$

Now,

$$\text{grad } u (\text{grad } v \times \text{grad } w) = \begin{vmatrix} 1 & 1 & 1 \\ 2x & 2y & 2z \\ z+y & z+x & y+x \end{vmatrix}$$

$$\begin{aligned}
&= 2 \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ z+y & z+x & y+x \end{vmatrix} \\
&= 2 \begin{vmatrix} 1 & 1 & 1 \\ x+z+y & y+z+x & z+y+x \\ z+y & z+x & y+x \end{vmatrix} \quad \text{Applying } R_2 \rightarrow R_2 + R_3 \\
&= 2(x+y+z) \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ y+z & z+x & x+y \end{vmatrix} = 0
\end{aligned}$$

Hence, grad u , grad v and grad w are coplanar vectors.

Example 14. Find the directional derivative of $\phi = 5x^2y - 5y^2z + \frac{5}{2}z^2x$ at the point $P(1, 1, 1)$ in the direction of the line $\frac{x-1}{2} = \frac{y-3}{-2} = \frac{z}{1}$. (U.P.T.U., 2003)

Sol. $\vec{\nabla}\phi = \left[10xy + \frac{5}{2}z^2\right]i + (5x^2 - 10yz)j + (-5y^2 + 5zx)k$

$$\nabla\phi \text{ at } P(1, 1, 1) = \frac{25}{2}i - 5j$$

Direction Ratio of the line $\frac{x-1}{2} = \frac{y-3}{-2} = \frac{z}{1}$ are 2, -2, 1

Direction cosines of the line are $\frac{2}{\sqrt{(2)^2 + (-2)^2 + (1)^2}}, \frac{-2}{\sqrt{2^2 + (-2)^2 + 1}}, \frac{1}{\sqrt{2^2 + (-2)^2 + 1}}$
i.e., $\frac{2}{3}, \frac{-2}{3}, \frac{1}{3}$

Directional derivative in the direction of the line

$$\begin{aligned}
&= \left(\frac{25}{2}i - 5j\right) \cdot \left(\frac{2}{3}i - \frac{2}{3}j + \frac{1}{3}k\right) \\
&= \frac{25}{3} + \frac{10}{3} \\
&= \frac{35}{3}.
\end{aligned}$$

Example 15. Prove that $\nabla \cdot \left\{ \frac{f(r)\vec{r}}{r} \right\} = \frac{1}{r^2} \frac{d}{dr}(r^2 f)$.

Sol.
$$\nabla \cdot \left\{ \frac{f(r)\vec{r}}{r} \right\} = \nabla \cdot \left\{ f(r) \frac{(x\hat{i} + y\hat{j} + z\hat{k})}{r} \right\}$$

$$= \frac{\partial}{\partial x} \left\{ \frac{f(r)x}{r} \right\} + \frac{\partial}{\partial y} \left\{ \frac{f(r)y}{r} \right\} + \frac{\partial}{\partial z} \left\{ \frac{f(r)z}{r} \right\}$$

Now,
$$\frac{\partial}{\partial x} \left\{ \frac{f(r)x}{r} \right\} = x \frac{d}{dr} \left\{ \frac{f(r)}{r} \right\} \frac{\partial r}{\partial x} + \frac{f(r)}{r}$$

$$= x \left\{ \frac{1}{r} \frac{df}{dr} - \frac{f(r)}{r^2} \right\} \frac{x}{r} + \frac{f(r)}{r} \quad \text{as} \quad \frac{\partial r}{\partial x} = \frac{x}{r}$$

$$= \frac{x^2}{r^2} \frac{df(r)}{dr} - \frac{x^2}{r^3} f(r) + \frac{f(r)}{r}$$

Similarly,
$$\frac{\partial}{\partial y} \left\{ \frac{f(r)y}{r} \right\} = \frac{y^2}{r^2} \frac{df(r)}{dr} - \frac{y^2}{r^3} f(r) + \frac{f(r)}{r}$$

and
$$\frac{\partial}{\partial z} \left\{ \frac{f(r)z}{r} \right\} = \frac{z^2}{r^2} \frac{df(r)}{dr} - \frac{z^2}{r^3} f(r) + \frac{f(r)}{r}$$

Now using these results, we get

$$\nabla \cdot \left[\frac{f(r)\vec{r}}{r} \right] = \frac{df(r)}{dr} + \frac{2}{r} f(r)$$

$$= \frac{1}{r^2} \frac{d}{dr} (r^2 f). \quad \text{Hence proved.}$$

Example 16. Find $f(r)$ such that $\nabla f = \frac{\vec{r}}{r^5}$ and $f(1) = 0$.

Sol. It is given that

$$\frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j + \frac{\partial f}{\partial z} k = \nabla f = \frac{\vec{r}}{r^5} = \frac{xi + yj + zk}{r^5}$$

So,
$$\frac{\partial f}{\partial x} = \frac{x}{r^5}, \quad \frac{\partial f}{\partial y} = \frac{y}{r^5} \quad \text{and} \quad \frac{\partial f}{\partial z} = \frac{z}{r^5}$$

We know that
$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = \frac{x}{r^5} dx + \frac{y}{r^5} dy + \frac{z}{r^5} dz$$

$$df = \frac{xdx + ydy + zdz}{r^5} = \frac{rdr}{r^5} = r^{-4} dr$$

Integrating
$$f(r) = \frac{r^{-3}}{-3} + c$$

Since
$$0 = f(1) = -\frac{1}{3} + c$$

So,
$$c = \frac{1}{3}$$

Thus,
$$f(r) = \frac{1}{3} - \frac{1}{3} \frac{1}{r^3}.$$

EXERCISE 5.2

1. Find grad f where $f = 2xz^4 - x^2y$ at $(2, -2, -1)$.

[Ans. $10i - 4j - 16k$]

2. Find ∇f when $f = (x^2 + y^2 + z^2) e^{-\sqrt{x^2 + y^2 + z^2}}$.

[Ans. $(2-r)e^{-r} \vec{r}$]

3. Find the unit normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$.

$$\left[\text{Ans. } \pm \frac{1}{3}(i - 2j - 2k) \right]$$

4. Find the directional derivative of $f = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2i - j - 2k$.

$$\left[\text{Ans. } \frac{37}{3} \right]$$

5. Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$.

$$\left[\text{Ans. } \theta = \cos^{-1} \left(\frac{8\sqrt{21}}{63} \right) \right]$$

6. Find the directional derivative of $f = xy + yz + zx$ in the direction of vector $i + 2j + 2k$ at the point $(1, 2, 0)$.

$$\left[\text{Ans. } \frac{10}{3} \right]$$

7. If $\nabla f = 2xyz^3i + x^2z^3j + 3x^2yz^2k$, find $f(x, y, z)$ if $f(1, -2, 2) = 4$. $\left[\text{Ans. } f = x^2yz^3 + 20 \right]$

8. Find f given $\nabla f = 2xi + 4yj + 8zk$. $\left[\text{Ans. } f = x^2 + 2y^2 + 4z^2 \right]$

9. Find the directional derivative of $\phi = (x^2 + y^2 + z^2)^{-1/2}$ at the point $P(3, 1, 2)$ in the direction of the vector, $yz i + xz j + xy k$.

$$\left[\text{Ans. } -\frac{9}{49\sqrt{14}} \right]$$

10. Prove that $\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$.

11. Show that $\nabla^2 \left(\frac{x}{r^3} \right) = 0$, where r is the magnitude of position vector $\vec{r} = xi + yj + zk$.

[U.P.T.U. (C.O.), 2002]

12. Find the direction in which the directional derivative of $f(x, y) = (x^2 - y^2)/xy$ at $(1, 1)$ is zero.

$$\left[\text{Ans. } \frac{1+i}{\sqrt{2}} \right]$$

13. Find the directional derivative of $\frac{1}{r}$ in the direction of $\vec{r}$, where $\vec{r} = xi + yj + zk$.

$$\left[\text{Ans. } -\frac{1}{r^2} \right] \text{ (U.P.T.U., 2003)}$$

14. Show that $\nabla r^{-3} = -3r^{-5} \vec{r}$.

15. If $\phi = \log |\vec{r}|$, show that $\nabla \phi = \frac{\vec{r}}{r^2}$.

[U.P.T.U., 2008]

5.10 DIVERGENCE OF A VECTOR POINT FUNCTION

If $\vec{f}(x, y, z)$ is any given continuously differentiable vector point function then the divergence of $\vec{f}$ scalar function defined as (U.P.T.U., 2006)

$$\nabla \cdot \vec{f} = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot \vec{f} = i \cdot \frac{\partial \vec{f}}{\partial x} + j \cdot \frac{\partial \vec{f}}{\partial y} + k \cdot \frac{\partial \vec{f}}{\partial z} = \text{div } \vec{f}$$

5.11 PHYSICAL INTERPRETATION OF DIVERGENCE

Let $\vec{v} = v_x i + v_y j + v_z k$ be the velocity of the fluid at $P(x, y, z)$.

Here we consider the case of fluid flow along a rectangular parallelopiped of dimensions $\delta x, \delta y, \delta z$

$$\text{Mass in} = v_x \delta y \delta z \quad (\text{along } x\text{-axis})$$

$$\begin{aligned} \text{Mass out} &= v_x(x + \delta x) \delta y \delta z \\ &= \left(v_x + \frac{\partial v_x}{\partial x} \delta x \right) \delta y \delta z \end{aligned}$$

| By Taylor's theorem

Net amount of mass along x -axis

$$\begin{aligned} &= v_x \delta y \delta z - \left(v_x + \frac{\partial v_x}{\partial x} \delta x \right) \delta y \delta z \\ &= - \frac{\partial v_x}{\partial x} \delta x \delta y \delta z \end{aligned}$$

| Minus sign shows decrease.

Similar net amount of mass along y -axis

$$= - \frac{\partial v_y}{\partial y} \delta x \delta y \delta z$$

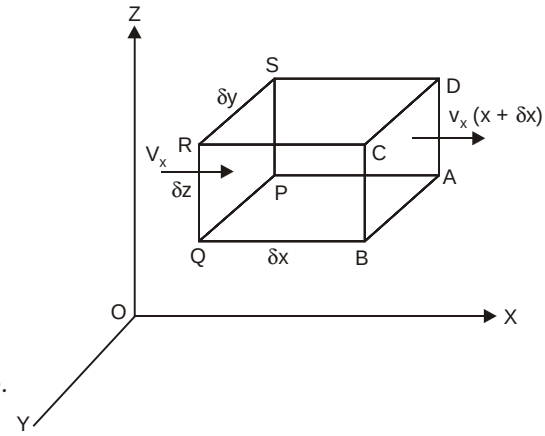


Fig. 5.4

$$\text{and net amount of mass along } z\text{-axis} = - \frac{\partial v_z}{\partial z} \delta x \delta y \delta z$$

$$\therefore \text{Total amount of fluid across parallelopiped per unit time} = - \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \delta x \delta y \delta z$$

Negative sign shows decrease of amount

$$\Rightarrow \text{Decrease of amount of fluid per unit time} = \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \delta x \delta y \delta z$$

Hence the rate of loss of fluid per unit volume

$$\begin{aligned} &= \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \\ &= \nabla \cdot \vec{v} = \text{div } \vec{v}. \end{aligned}$$

Therefore, $\text{div } \vec{v}$ represents the rate of loss of fluid per unit volume.

Solenoidal: For compressible fluid there is no gain no loss in the volume element

$$\therefore \operatorname{div} \vec{v} = 0$$

then $\vec{v}$ is called Solenoidal vector function.

5.12 CURL OF A VECTOR

If $\vec{f}$ is any given continuously differentiable vector point function then the curl of $\vec{f}$ (vector function) is defined as

$$\operatorname{Curl} \vec{f} = \nabla \times \vec{f} = i \times \frac{\partial \vec{f}}{\partial x} + j \times \frac{\partial \vec{f}}{\partial y} + k \times \frac{\partial \vec{f}}{\partial z} \quad (\text{U.P.T.U., 2006})$$

Let

$$\vec{f} = f_x i + f_y j + f_z k, \text{ then}$$

$$\nabla \times \vec{f} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f_x & f_y & f_z \end{vmatrix}.$$

5.13 PHYSICAL MEANING OF CURL

Here we consider the relation $\vec{v} = \vec{\omega} \times \vec{r}$, $\vec{\omega}$ is the angular velocity $\vec{r}$ is position vector of a point on the rotating body (U.P.T.U., 2006)

$$\begin{aligned} \operatorname{curl} \vec{v} &= \nabla \times \vec{v} \\ &= \nabla \times (\vec{\omega} \times \vec{r}) \\ &= \nabla \times [(w_1 i + w_2 j + w_3 k) \times (xi + yj + zk)] & \left[\begin{array}{l} \vec{\omega} = w_1 i + w_2 j + w_3 k \\ \vec{r} = xi + yj + zk \end{array} \right] \\ &= \nabla \times \begin{vmatrix} i & j & k \\ w_1 & w_2 & w_3 \\ x & y & z \end{vmatrix} \\ &= \nabla \times [(w_2 z - w_3 y)i - (w_1 z - w_3 x)j + (w_1 y - w_2 x)k] \\ &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \times [(w_2 z - w_3 y)i - (w_1 z - w_3 x)j + (w_1 y - w_2 x)k] \\ &= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ w_2 z - w_3 y & w_3 x - w_1 z & w_1 y - w_2 x \end{vmatrix} \\ &= (w_1 + w_1) i - (-w_2 - w_2) j + (w_3 + w_3) k \\ &= 2(w_1 i + w_2 j + w_3 k) = 2\vec{\omega} \end{aligned}$$

$\operatorname{Curl} \vec{v} = 2\vec{\omega}$ which shows that curl of a vector field is connected with rotational properties of the vector field and justifies the name rotation used for curl.

Irrotational vector: If $\text{curl } \vec{f} = 0$, then the vector $\vec{f}$ is said to be irrotational. Vice-versa, if $\vec{f}$ is irrotational then, $\text{curl } \vec{f} = 0$.

5.14 VECTOR IDENTITIES

Identity 1: $\text{grad } uv = u \text{ grad } v + v \text{ grad } u$

Proof:
$$\begin{aligned}\text{grad } (uv) &= \nabla (uv) \\ &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (uv) \\ &= \hat{i} \frac{\partial}{\partial x} (uv) + \hat{j} \frac{\partial}{\partial y} (uv) + \hat{k} \frac{\partial}{\partial z} (uv) \\ &= i \left(u \frac{\partial v}{\partial x} + v \frac{\partial u}{\partial x} \right) + j \left(u \frac{\partial v}{\partial y} + v \frac{\partial u}{\partial y} \right) + k \left(u \frac{\partial v}{\partial z} + v \frac{\partial u}{\partial z} \right) \\ &= u \left(i \frac{\partial v}{\partial x} + j \frac{\partial v}{\partial y} + k \frac{\partial v}{\partial z} \right) + v \left(i \frac{\partial u}{\partial x} + j \frac{\partial u}{\partial y} + k \frac{\partial u}{\partial z} \right)\end{aligned}$$

or

$$\boxed{\text{grad } uv = u \text{ grad } v + v \text{ grad } u}.$$

Identity 2: $\text{grad } (\vec{a} \cdot \vec{b}) = \vec{a} \times \text{curl } \vec{b} + \vec{b} \times \text{curl } \vec{a} + (\vec{a} \cdot \nabla) \vec{b} + (\vec{b} \cdot \nabla) \vec{a}$

Proof:
$$\begin{aligned}\text{grad } (\vec{a} \cdot \vec{b}) &= \Sigma i \frac{\partial}{\partial x} (\vec{a} \cdot \vec{b}) = \Sigma i \left(\frac{\partial \vec{a}}{\partial x} \cdot \vec{b} + \vec{a} \cdot \frac{\partial \vec{b}}{\partial x} \right) \\ &= \Sigma i \left(\vec{b} \cdot \frac{\partial \vec{a}}{\partial x} \right) + \Sigma i \left(\vec{a} \cdot \frac{\partial \vec{b}}{\partial x} \right).\end{aligned}\tag{... (i)}$$

Now,
$$\vec{a} \times \left(i \times \frac{\partial \vec{b}}{\partial x} \right) = \left(\vec{a} \cdot \frac{\partial \vec{b}}{\partial x} \right) i - (\vec{a} \cdot i) \frac{\partial \vec{b}}{\partial x}$$

$$\Rightarrow \left(\vec{a} \cdot \frac{\partial \vec{b}}{\partial x} \right) i = \vec{a} \times \left(i \times \frac{\partial \vec{b}}{\partial x} \right) + (\vec{a} \cdot i) \frac{\partial \vec{b}}{\partial x}$$

$$\Rightarrow \Sigma \left(\vec{a} \cdot \frac{\partial \vec{b}}{\partial x} \right) i = \Sigma \vec{a} \times \left(i \times \frac{\partial \vec{b}}{\partial x} \right) + \Sigma (\vec{a} \cdot i) \frac{\partial \vec{b}}{\partial x}$$

$$\Rightarrow \Sigma \left(\vec{a} \cdot \frac{\partial \vec{b}}{\partial x} \right) i = \vec{a} \times \Sigma \left(i \times \frac{\partial \vec{b}}{\partial x} \right) + \Sigma \left(\vec{a} \cdot i \frac{\partial}{\partial x} \right) \vec{b}$$

$$\Rightarrow \Sigma \left(\vec{a} \cdot \frac{\partial \vec{b}}{\partial x} \right) i = \vec{a} \times \text{curl } \vec{b} + (\vec{a} \cdot \nabla) \vec{b}.\tag{... (ii)}$$

Interchanging $\vec{a}$ and $\vec{b}$, we get

$$\Sigma \left(\vec{b} \cdot \frac{\partial \vec{a}}{\partial x} \right) i = \vec{b} \times \text{curl } \vec{a} + (\vec{b} \cdot \nabla) \vec{a}\tag{... (iii)}$$

From equations (i), (ii) and (iii), we get

$$\boxed{\text{grad } (\vec{a} \cdot \vec{b}) = \vec{a} \times \text{curl } \vec{b} + \vec{b} \times \text{curl } \vec{a} + (\vec{a} \cdot \nabla) \vec{b} + (\vec{b} \cdot \nabla) \vec{a}}.$$

Identity 3: $\operatorname{div} (u \vec{a}) = u \operatorname{div} \vec{a} + \vec{a} \cdot \operatorname{grad} u$ (U.P.T.U., 2004)

Proof:
$$\begin{aligned} \operatorname{div} (u \vec{a}) &= \nabla \cdot (u \vec{a}) \\ &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (u \vec{a}) \\ &= i \cdot \frac{\partial}{\partial x} (u \vec{a}) + j \cdot \frac{\partial}{\partial y} (u \vec{a}) + k \cdot \frac{\partial}{\partial z} (u \vec{a}) \\ &= i \cdot \left\{ \frac{\partial u}{\partial x} \vec{a} + u \frac{\partial \vec{a}}{\partial x} \right\} + j \cdot \left\{ \frac{\partial u}{\partial y} \vec{a} + u \frac{\partial \vec{a}}{\partial y} \right\} + k \cdot \left\{ \frac{\partial u}{\partial z} \vec{a} + u \frac{\partial \vec{a}}{\partial z} \right\} \\ &= u \left\{ i \cdot \frac{\partial \vec{a}}{\partial x} + j \cdot \frac{\partial \vec{a}}{\partial y} + k \cdot \frac{\partial \vec{a}}{\partial z} \right\} + \vec{a} \cdot \left\{ i \frac{\partial u}{\partial x} + j \frac{\partial u}{\partial y} + k \frac{\partial u}{\partial z} \right\} \end{aligned}$$

or

$$\boxed{\operatorname{div} (u \vec{a}) = u \operatorname{div} \vec{a} + \vec{a} \cdot \operatorname{grad} u}.$$

Identity 4: $\operatorname{div} (\vec{a} \times \vec{b}) = \vec{b} \cdot \operatorname{curl} \vec{a} - \vec{a} \cdot \operatorname{curl} \vec{b}$ [U.P.T.U. (C.O.), 2003]

Proof:
$$\begin{aligned} \operatorname{div} (\vec{a} \times \vec{b}) &= \nabla \cdot (\vec{a} \times \vec{b}) \\ &= \Sigma i \cdot \frac{\partial}{\partial x} (\vec{a} \times \vec{b}) \\ &= \Sigma i \cdot \left(\frac{\partial \vec{a}}{\partial x} \times \vec{b} + \vec{a} \times \frac{\partial \vec{b}}{\partial x} \right) \\ &= \Sigma i \cdot \left(\frac{\partial \vec{a}}{\partial x} \times \vec{b} \right) + \Sigma i \cdot \left(\vec{a} \times \frac{\partial \vec{b}}{\partial x} \right) \\ &= \Sigma \left(i \times \frac{\partial \vec{a}}{\partial x} \right) \cdot \vec{b} - \Sigma \left(i \times \frac{\partial \vec{b}}{\partial x} \right) \cdot \vec{a} \\ &= (\operatorname{curl} \vec{a}) \cdot \vec{b} - (\operatorname{curl} \vec{b}) \cdot \vec{a} \end{aligned}$$

or

$$\boxed{\operatorname{div} (\vec{a} \times \vec{b}) = \vec{b} \cdot \operatorname{curl} \vec{a} - \vec{a} \cdot \operatorname{curl} \vec{b}}.$$

Identity 5: $\operatorname{curl} (u \vec{a}) = u \operatorname{curl} \vec{a} + (\operatorname{grad} u) \times \vec{a}$ [U.P.T.U. (C.O.), 2003]

Proof:
$$\begin{aligned} \operatorname{curl} (u \vec{a}) &= \nabla \times (u \vec{a}) \\ &= \Sigma i \times \frac{\partial}{\partial x} (u \vec{a}) \\ &= \Sigma i \times \left(\frac{\partial u}{\partial x} \vec{a} + u \frac{\partial \vec{a}}{\partial x} \right) \\ &= \Sigma \left(i \frac{\partial u}{\partial x} \right) \times \vec{a} + u \Sigma \left(i \times \frac{\partial \vec{a}}{\partial x} \right) \\ &= (\operatorname{grad} u) \times \vec{a} + u \operatorname{curl} \vec{a} \end{aligned}$$

or

$$\boxed{\operatorname{curl} (u \vec{a}) = u \operatorname{curl} \vec{a} + (\operatorname{grad} u) \times \vec{a}}.$$

Identity 6: $\text{curl}(\vec{a} \times \vec{b}) = \vec{a} \text{ div } \vec{b} - \vec{b} \text{ div } \vec{a} + (\vec{b} \cdot \nabla) \vec{a} - (\vec{a} \cdot \nabla) \vec{b}$

Proof:
$$\begin{aligned} \text{curl}(\vec{a} \times \vec{b}) &= \nabla \times (\vec{a} \times \vec{b}) \\ &= \Sigma i \times \frac{\partial}{\partial x} (\vec{a} \times \vec{b}) \\ &= \Sigma i \times \left(\frac{\partial \vec{a}}{\partial x} \times \vec{b} + \vec{a} \times \frac{\partial \vec{b}}{\partial x} \right) \\ &= \Sigma i \times \left(\frac{\partial \vec{a}}{\partial x} \times \vec{b} \right) + \Sigma i \times \left(\vec{a} \times \frac{\partial \vec{b}}{\partial x} \right) \\ &= \Sigma (i \cdot b) \frac{\partial \vec{a}}{\partial x} - \Sigma \left(i \cdot \frac{\partial \vec{a}}{\partial x} \right) \vec{b} + \Sigma \left(i \cdot \frac{\partial \vec{b}}{\partial x} \right) \vec{a} - \Sigma (i \cdot \vec{a}) \frac{\partial \vec{b}}{\partial x} \\ &= \Sigma (i \cdot \vec{b}) \frac{\partial}{\partial x} \vec{a} - \Sigma \left(i \cdot \frac{\partial \vec{a}}{\partial x} \right) \vec{b} + \Sigma \left(i \cdot \frac{\partial \vec{b}}{\partial x} \right) \vec{a} - \Sigma \left(\vec{a} \cdot i \frac{\partial}{\partial x} \right) \vec{b} \\ &= (\vec{b} \cdot \nabla) \vec{a} - (\text{div } \vec{a}) \vec{b} + (\text{div } \vec{b}) \vec{a} - (\vec{a} \cdot \nabla) \vec{b} \\ &= (\vec{b} \cdot \nabla) \vec{a} - (\vec{a} \cdot \nabla) \vec{b} + \vec{a} \text{ div } \vec{b} - \vec{b} \text{ div } \vec{a}. \end{aligned}$$

or

$$\text{curl}(\vec{a} \times \vec{b}) = \vec{a} \text{ div } \vec{b} - \vec{b} \text{ div } \vec{a} + (\vec{b} \cdot \nabla) \vec{a} - (\vec{a} \cdot \nabla) \vec{b}.$$

Identity 7: $\text{div grad } f = \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \nabla^2 f$

Proof:
$$\begin{aligned} \text{div grad } f &= \nabla \cdot (\nabla f) \\ &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot \left(i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial z} \right) \\ &= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \end{aligned}$$

or

$$\text{div grad } f = \nabla^2 f.$$

Identity 8: $\text{curl grad } f = 0$

Proof:
$$\begin{aligned} \text{curl grad } f &= \nabla \times (\nabla f) \\ &= \Sigma i \frac{\partial}{\partial x} \times \left(i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z} \right) \\ &= \Sigma i \times \left(i \frac{\partial^2 f}{\partial x^2} + j \frac{\partial^2 f}{\partial x \partial y} + k \frac{\partial^2 f}{\partial x \partial z} \right) \\ &= \Sigma \left(k \frac{\partial^2 f}{\partial x \partial y} - j \frac{\partial^2 f}{\partial x \partial z} \right) \\ &= \left(k \frac{\partial^2 f}{\partial x \partial y} - j \frac{\partial^2 f}{\partial x \partial z} \right) + \left(i \frac{\partial^2 f}{\partial y \partial z} - k \frac{\partial^2 f}{\partial y \partial x} \right) + \left(j \frac{\partial^2 f}{\partial z \partial x} - i \frac{\partial^2 f}{\partial z \partial y} \right) \end{aligned}$$

or

$$\text{curl grad } f = 0.$$

Identity 9: $\text{div curl } \vec{f} = 0$

Proof:
$$\begin{aligned}\text{div curl } \vec{f} &= \nabla \cdot (\nabla \times \vec{f}) \\ &= \Sigma i \cdot \frac{\partial}{\partial x} \left\{ i \times \frac{\partial \vec{f}}{\partial x} + j \times \frac{\partial \vec{f}}{\partial y} + k \times \frac{\partial \vec{f}}{\partial z} \right\} \\ &= \Sigma i \cdot \left\{ i \times \frac{\partial^2 \vec{f}}{\partial x^2} + j \times \frac{\partial^2 \vec{f}}{\partial x \partial y} + k \times \frac{\partial^2 \vec{f}}{\partial x \partial z} \right\} \\ &= \Sigma \left\{ (i \times i) \cdot \frac{\partial^2 \vec{f}}{\partial x^2} + (i \times j) \cdot \frac{\partial^2 \vec{f}}{\partial x \partial y} + (i \times k) \cdot \frac{\partial^2 \vec{f}}{\partial x \partial z} \right\} \\ &= \Sigma \left\{ k \cdot \frac{\partial^2 \vec{f}}{\partial x \partial y} - j \cdot \frac{\partial^2 \vec{f}}{\partial x \partial z} \right\} \\ &= \left\{ k \cdot \frac{\partial^2 \vec{f}}{\partial x \partial y} - j \cdot \frac{\partial^2 \vec{f}}{\partial x \partial z} \right\} + \left\{ i \cdot \frac{\partial^2 \vec{f}}{\partial y \partial z} - k \cdot \frac{\partial^2 \vec{f}}{\partial y \partial x} \right\} + \left\{ j \cdot \frac{\partial^2 \vec{f}}{\partial z \partial x} - i \cdot \frac{\partial^2 \vec{f}}{\partial z \partial y} \right\} \\ &\quad \boxed{\text{div curl } \vec{f} = 0}.\end{aligned}$$

Identity 10: $\text{grad div } \vec{f} = \text{curl curl } \vec{f} + \frac{\partial^2 \vec{f}}{\partial x^2} + \frac{\partial^2 \vec{f}}{\partial y^2} + \frac{\partial^2 \vec{f}}{\partial z^2}$

Proof:
$$\begin{aligned}\text{Curl curl } \vec{f} &= \nabla \times (\nabla \times \vec{f}) \\ &= \Sigma i \frac{\partial}{\partial x} \times \left\{ i \times \frac{\partial \vec{f}}{\partial x} + j \times \frac{\partial \vec{f}}{\partial y} + k \times \frac{\partial \vec{f}}{\partial z} \right\} \\ &= \Sigma i \times \left\{ i \times \frac{\partial^2 \vec{f}}{\partial x^2} + j \times \frac{\partial^2 \vec{f}}{\partial x \partial y} + k \times \frac{\partial^2 \vec{f}}{\partial x \partial z} \right\} \\ &= \Sigma \left[\left\{ \left(i \cdot \frac{\partial^2 \vec{f}}{\partial x^2} \right) i - (i \cdot i) \cdot \frac{\partial^2 \vec{f}}{\partial x^2} \right\} + \left\{ \left(i \cdot \frac{\partial^2 \vec{f}}{\partial x \partial y} \right) j - (i \cdot j) \frac{\partial^2 \vec{f}}{\partial x \partial y} \right\} \right. \\ &\quad \left. + \left\{ \left(i \cdot \frac{\partial^2 \vec{f}}{\partial x \partial z} \right) k - (i \cdot k) \frac{\partial^2 \vec{f}}{\partial x \partial z} \right\} \right] \\ &= \Sigma \left[\left(i \cdot \frac{\partial^2 \vec{f}}{\partial x^2} \right) i + \left(i \cdot \frac{\partial^2 \vec{f}}{\partial x \partial y} \right) j + \left(i \cdot \frac{\partial^2 \vec{f}}{\partial x \partial z} \right) k \right] - \Sigma \frac{\partial^2 \vec{f}}{\partial x^2} \\ \Rightarrow \text{Curl curl } \vec{f} + \frac{\partial^2 \vec{f}}{\partial x^2} + \frac{\partial^2 \vec{f}}{\partial y^2} + \frac{\partial^2 \vec{f}}{\partial z^2} &= \Sigma \left[\left(i \cdot \frac{\partial^2 \vec{f}}{\partial x^2} \right) i + \left(i \cdot \frac{\partial^2 \vec{f}}{\partial x \partial y} \right) j + \left(i \cdot \frac{\partial^2 \vec{f}}{\partial x \partial z} \right) k \right]. \quad \dots(i)\end{aligned}$$

Again,

$$\begin{aligned}\text{grad div } \vec{f} &= \Sigma i \frac{\partial}{\partial x} \left\{ i \cdot \frac{\partial \vec{f}}{\partial x} + j \cdot \frac{\partial \vec{f}}{\partial y} + k \cdot \frac{\partial \vec{f}}{\partial z} \right\} \\ &= \Sigma i \left\{ i \cdot \frac{\partial^2 \vec{f}}{\partial x^2} + j \cdot \frac{\partial^2 \vec{f}}{\partial x \partial y} + k \cdot \frac{\partial^2 \vec{f}}{\partial x \partial z} \right\}\end{aligned}$$

$$\begin{aligned}
&= \Sigma \left[\left(\hat{i} \cdot \frac{\partial^2 \vec{f}}{\partial x^2} \right) \hat{i} + \left(\hat{j} \cdot \frac{\partial^2 \vec{f}}{\partial x \partial y} \right) \hat{i} + \left(\hat{k} \cdot \frac{\partial^2 \vec{f}}{\partial x \partial z} \right) \hat{i} \right] \\
&= \Sigma \left[\left(\hat{i} \cdot \frac{\partial^2 \vec{f}}{\partial x^2} \right) \hat{i} + \left(\hat{i} \cdot \frac{\partial^2 \vec{f}}{\partial z \partial x} \right) \hat{j} + \left(\hat{i} \cdot \frac{\partial^2 \vec{f}}{\partial y \partial z} \right) \hat{k} \right] \quad \dots(ii)
\end{aligned}$$

From eqns. (i) and (ii), we prove

$$\text{grad div } \vec{f} = \text{curl curl } \vec{f} + \nabla^2 \vec{f}$$

Example 1. If $\vec{f} = xy^2 \hat{i} + 2x^2yz \hat{j} - 3yz^2 \hat{k}$ then find $\text{div } \vec{f}$ and $\text{curl } \vec{f}$ at the point $(1, -1, 1)$.

Sol. We have $\vec{f} = xy^2 \hat{i} + 2x^2yz \hat{j} - 3yz^2 \hat{k}$

$$\begin{aligned}
\text{div } \vec{f} &= \frac{\partial}{\partial x}(xy^2) + \frac{\partial}{\partial y}(2x^2yz) + \frac{\partial}{\partial z}(-3yz^2) \\
&= y^2 + 2x^2z - 6yz \\
&= (-1)^2 + 2(1)^2(1) - 6(-1)(1) \text{ at } (1, -1, 1) \\
&= 1 + 2 + 6 = 9.
\end{aligned}$$

Again, $\text{curl } \vec{f} = \text{curl } [xy^2 \hat{i} + 2x^2yz \hat{j} - 3yz^2 \hat{k}]$

$$\begin{aligned}
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2 & 2x^2yz & -3yz^2 \end{vmatrix} \\
&= \hat{i} \left\{ \frac{\partial}{\partial y}(-3yz^2) - \frac{\partial}{\partial z}(2x^2yz) \right\} + \hat{j} \left\{ \frac{\partial}{\partial z}(xy^2) - \frac{\partial}{\partial x}(-3yz^2) \right\} \\
&\quad + \hat{k} \left\{ \frac{\partial}{\partial x}(2x^2yz) - \frac{\partial}{\partial y}(xy^2) \right\} \\
&= \hat{i} [-3z^2 - 2x^2y] + \hat{j} [0 - 0] + \hat{k} [4xyz - 2xy] \\
&= (-3z^2 - 2x^2y)\hat{i} + (4xyz - 2xy)\hat{k} \\
&= \{-3(1)^2 - 2(1)^2(-1)\} \hat{i} + \{4(1)(-1)(1) - 2(1)(-1)\} \hat{k} \text{ at } (1, -1, 1) \\
&= -\hat{i} - 2\hat{k}.
\end{aligned}$$

Example 2. Prove that

$$(i) \text{ div } \vec{r} = 3. \quad (ii) \text{ curl } \vec{r} = 0.$$

Sol. (i) $\text{div } \vec{r} = \nabla \cdot \vec{r}$

$$\begin{aligned}
&= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (xi + yj + zk) \\
&= \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) \\
&= 1 + 1 + 1 = 3.
\end{aligned}$$

(ii) $\text{curl } \vec{r} = \nabla \times \vec{r}$

$$\begin{aligned}
&= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times (xi + yj + zk) \\
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix}
\end{aligned}$$

$$\begin{aligned}
&= i \left[\frac{\partial}{\partial y}(z) - \frac{\partial}{\partial z}(y) \right] + j \left[\frac{\partial}{\partial z}(x) - \frac{\partial}{\partial x}(z) \right] + k \left[\frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(x) \right] \\
&= i(0) + j(0) + k(0) \\
&= 0 + 0 + 0 = 0.
\end{aligned}$$

Example 3. Find the divergence and curl of the vector

$$(x^2 - y^2) i + 2xy j + (y^2 - xy) k.$$

Sol. Let $\vec{f} = (x^2 - y^2) i + 2xy j + (y^2 - xy) k$.

$$\begin{aligned}
\text{Then } \operatorname{div} \vec{f} &= \frac{\partial}{\partial x}(x^2 - y^2) + \frac{\partial}{\partial y}(2xy) + \frac{\partial}{\partial z}(y^2 - xy) \\
&= 2x + 2x + 0 = 4x
\end{aligned}$$

$$\begin{aligned}
\text{and } \operatorname{curl} \vec{f} &= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - y^2 & 2xy & y^2 - xy \end{vmatrix} \\
&= i \left[\frac{\partial}{\partial y}(y^2 - xy) - \frac{\partial}{\partial z}(2xy) \right] + j \left[\frac{\partial}{\partial z}(x^2 - y^2) - \frac{\partial}{\partial x}(y^2 - xy) \right] \\
&\quad + k \left[\frac{\partial}{\partial x}(2xy) - \frac{\partial}{\partial y}(x^2 - y^2) \right] \\
&= i [2y - x - 0] + j [0 - (-y)] + k [(2y) - (-2y)] \\
&= (2y - x) i + y j + 4y k. \quad \text{Ans.}
\end{aligned}$$

Example 4. If $\vec{f}(x, y, z) = xz^3 \hat{i} - 2x^2yz \hat{j} + 2yz^4 \hat{k}$ find divergence and curl of $\vec{f}(x, y, z)$
(U.P.T.U., 2006)

$$\begin{aligned}
\text{Sol. } \operatorname{div} \vec{f} &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (xz^3 i - 2x^2yz j + 2yz^4 k) \\
&= \frac{\partial}{\partial x}(xz^3) - \frac{\partial}{\partial y}(2x^2yz) + \frac{\partial}{\partial z}(2yz^4) \\
&= z^3 - 2x^2z + 8yz^3 \\
\operatorname{curl} \vec{f} &= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix} \\
&= i \left\{ \frac{\partial}{\partial y}(2yz^4) + \frac{\partial}{\partial z}(2x^2yz) \right\} - j \left\{ \frac{\partial}{\partial x}(2yz^4) - \frac{\partial}{\partial z}(xz^3) \right\} \\
&\quad + k \left\{ \frac{\partial}{\partial x}(-2x^2yz) - \frac{\partial}{\partial y}(xz^3) \right\} \\
&= i (2z^4 + 2x^2y) - j (0 - 3z^2x) + k (-4xyz - 0) \\
&= 2(x^2y + z^4) i + 3z^2x j - 4xyz \cdot k.
\end{aligned}$$

Example 5. Find the directional derivative of $\nabla \cdot \vec{u}$ at the point $(4, 4, 2)$ in the direction of the corresponding outer normal of the sphere $x^2 + y^2 + z^2 = 36$ where $\vec{u} = x^4 \vec{i} + y^4 \vec{j} + z^4 \vec{k}$.

Sol. $\nabla \cdot \vec{u} = \nabla \cdot (x^4 \vec{i} + y^4 \vec{j} + z^4 \vec{k}) = 4(x^3 + y^3 + z^3) = f$ (say)

$\therefore (\nabla f)_{(4, 4, 2)} = 12(x^2 \vec{i} + y^2 \vec{j} + z^2 \vec{k})_{(4, 4, 2)} = 48(4\vec{i} + 4\vec{j} + \vec{k})$

Normal to the sphere $g \equiv x^2 + y^2 + z^2 = 36$ is

$$(\nabla g)_{(4, 4, 2)} = 2(x\vec{i} + y\vec{j} + z\vec{k})_{(4, 4, 2)} = 4(2\vec{i} + 2\vec{j} + \vec{k})$$

$$\begin{aligned} \hat{a} &= \text{unit normal} = \frac{\nabla g}{|\nabla g|} = \frac{4(2\vec{i} + 2\vec{j} + \vec{k})}{\sqrt{64 + 64 + 16}} \\ &= \frac{2\vec{i} + 2\vec{j} + \vec{k}}{3} \end{aligned}$$

The required directional derivative is

$$\begin{aligned} \nabla f \cdot \hat{a} &= 48(4\vec{i} + 4\vec{j} + \vec{k}) \cdot \frac{2\vec{i} + 2\vec{j} + \vec{k}}{3} \\ &= 16(8 + 8 + 1) = 272. \end{aligned}$$

Example 6. A fluid motion is given by $\vec{v} = (y + z)\vec{i} + (z + x)\vec{j} + (x + y)\vec{k}$ show that the motion is irrotational and hence find the scalar potential. (U.P.T.U., 2003)

Sol. $\text{Curl } \vec{v} = \Delta \times \vec{v}$

$$\begin{aligned} &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \times [(y + z)\vec{i} + (z + x)\vec{j} + (x + y)\vec{k}] \\ &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y + z & z + x & x + y \end{vmatrix} = \vec{i}(1 - 1) - \vec{j}(1 - 1) + \vec{k}(1 - 1) = 0 \end{aligned}$$

Hence $\vec{v}$ is irrotational.

Now

$$\begin{aligned} d\phi &= \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz \\ &= \left(\vec{i} \frac{\partial \phi}{\partial x} + \vec{j} \frac{\partial \phi}{\partial y} + \vec{k} \frac{\partial \phi}{\partial z} \right) \cdot (dx + dy + dz) \\ &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \phi \cdot d\vec{r} = \nabla \phi \cdot d\vec{r} = \vec{v} \cdot d\vec{r} \quad \left| \vec{v} = \nabla \phi \right. \\ &= [(y + z)\vec{i} + (z + x)\vec{j} + (x + y)\vec{k}] \cdot (dx + dy + dz) \\ &= (y + z) dx + (z + x) dy + (x + y) dz \\ &= ydx + zdx + zdy + xdy + xdz + ydz \end{aligned}$$

On integrating

$$\begin{aligned} \phi &= \int (ydx + xdy) + \int (zdy + ydz) + \int (zdx + xdz) \\ &= \int d(xy) + \int d(yz) + \int d(zx) \end{aligned}$$

$$\phi = xy + yz + zx + c$$

Thus, velocity potential = $xy + yz + zx + c$.

Example 7. Prove that $\vec{a} \times (\nabla \times \vec{r}) = \nabla(\vec{a} \cdot \vec{r}) - (\vec{a} \cdot \nabla) \vec{r}$ where $\vec{a}$ is a constant vector and $\vec{r} = xi + yj + zk$. (U.P.T.U., 2007)

Sol. Let

$$\begin{aligned}\vec{a} &= a_1i + a_2j + a_3k \\ \vec{r} &= r_1i + r_2j + r_3k\end{aligned}$$

$$\therefore \nabla \times \vec{r} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ r_1 & r_2 & r_3 \end{vmatrix} = i \left(\frac{\partial r_3}{\partial y} - \frac{\partial r_2}{\partial z} \right) - \left(\frac{\partial r_3}{\partial x} - \frac{\partial r_1}{\partial z} \right) j + \left(\frac{\partial r_2}{\partial x} - \frac{\partial r_1}{\partial y} \right) k$$

$$\begin{aligned}\text{Now } \vec{a} \times (\nabla \times \vec{r}) &= \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ \frac{\partial r_3}{\partial y} - \frac{\partial r_2}{\partial z} & \frac{\partial r_1}{\partial z} - \frac{\partial r_3}{\partial x} & \frac{\partial r_2}{\partial x} - \frac{\partial r_1}{\partial y} \end{vmatrix} \\ &= \left\{ \left(a_2 \frac{\partial r_2}{\partial x} - a_2 \frac{\partial r_1}{\partial y} \right) - \left(a_3 \frac{\partial r_1}{\partial z} - a_3 \frac{\partial r_3}{\partial x} \right) \right\} i \\ &\quad - \left\{ \left(a_1 \frac{\partial r_2}{\partial x} - a_1 \frac{\partial r_1}{\partial y} \right) - \left(a_3 \frac{\partial r_3}{\partial y} - a_3 \frac{\partial r_2}{\partial z} \right) \right\} j \\ &\quad + \left\{ \left(-a_1 \frac{\partial r_3}{\partial x} + a_1 \frac{\partial r_1}{\partial z} \right) - \left(a_2 \frac{\partial r_3}{\partial y} - a_2 \frac{\partial r_2}{\partial z} \right) \right\} k \\ &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (a_1 r_1 + a_2 r_2 + a_3 r_3) - \left[a_1 \frac{\partial}{\partial x} + a_2 \frac{\partial}{\partial y} + a_3 \frac{\partial}{\partial z} \right] (r_1 i + r_2 j + r_3 k) \\ &= \nabla \{ (a_1 i + a_2 j + a_3 k) \cdot (r_1 i + r_2 j + r_3 k) \} \\ &\quad - \left\{ (a_1 i + a_2 j + a_3 k) \cdot \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \right\} (r_1 i + r_2 j + r_3 k) \\ &= \nabla(\vec{a} \cdot \vec{r}) - (\vec{a} \cdot \nabla) \vec{r}. \text{ Hence proved.}\end{aligned}$$

Example 8. Find the directional derivative of $\nabla \cdot (\nabla f)$ at the point $(1, -2, 1)$ in the direction of the normal to the surface $xy^2z = 3x + z^2$ where $f = 2x^3y^2z^4$. (U.P.T.U., 2008)

Sol.
$$\nabla f = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (2x^3y^2z^4) = (6x^2y^2z^4)i + (4x^3yz^4)j + 8x^3y^2z^3k$$

$$\nabla \cdot (\nabla f) = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot \{ (6x^2y^2z^4)i + (4x^3yz^4)j + (8x^3y^2z^3)k \}$$

or

$$\nabla \cdot (\nabla f) = 12x y^2 z^4 + 4x^3 z^4 + 24x^3 y^2 z^2 = F(x, y, z) \text{ (say)}$$

Now

$$\begin{aligned}\nabla F &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (12x y^2 z^4 + 4x^3 z^4 + 24x^3 y^2 z^2) \\ &= (12y^2 z^4 + 12x^2 z^4 + 72x^2 y^2 z^2)i + (24xyz^4 + 48x^3 yz^2)j \\ &\quad + (48x y^2 z^3 + 16x^3 z^3 + 48x^3 y^2 z)k\end{aligned}$$

$$\begin{aligned}
 \therefore (\nabla F)_{(1, -2, 1)} &= 348i - 144j + 400k \\
 \text{Let } g(x, y, z) &= xy^2z - 3x - z^2 = 0 \\
 \nabla g &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (xy^2z - 3x - z^2) \\
 &= (y^2z - 3)i + (2xyz)j + (xy^2 - 2z)k \\
 (\nabla g)_{(1, -2, 1)} &= i - 4j + 2k \\
 \hat{a} = \text{unit normal} &= \frac{\nabla g}{|\nabla g|} = \frac{i - 4j + 2k}{\sqrt{1 + 16 + 4}} = \frac{i - 4j + 2k}{\sqrt{21}}
 \end{aligned}$$

Hence, the required directional derivative is

$$\begin{aligned}
 \nabla F \cdot \hat{a} &= (348i - 144j + 400k) \cdot \frac{(i - 4j + 2k)}{\sqrt{21}} \\
 &= \frac{348 + 576 + 800}{\sqrt{21}} = \frac{1724}{\sqrt{21}}.
 \end{aligned}$$

Example 9. Determine the values of a and b so that the surface $ax^2 - byz = (a + 2)x$ will be orthogonal to the surface $4x^2y + z^3 = 4$ at the point $(1, -1, 2)$.

Sol. Let $f \equiv ax^2 - byz - (a + 2)x = 0$... (i)

$g \equiv 4x^2y + z^3 - 4 = 0$... (ii)

$$\begin{aligned}
 \text{grad } f = \nabla f &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \{ax^2 - byz - (a + 2)x\} \\
 &= (2ax - a - 2)i + (-bz)j + (-by)k \\
 (\nabla f)_{(1, -1, 2)} &= (a - 2)i - 2bj + bk \quad \dots (iii)
 \end{aligned}$$

$$\begin{aligned}
 \text{grad } g = \nabla g &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (4x^2y + z^3 - 4) \\
 &= (8xy)i + (4x^2)j + (3z^2)k \\
 (\nabla g)_{(1, -1, 2)} &= -8i + 4j + 12k \quad \dots (iv)
 \end{aligned}$$

Since the surfaces are orthogonal so

$$\begin{aligned}
 (\vec{\nabla} f) \cdot (\vec{\nabla} g) &= 0 \\
 \Rightarrow [(a - 2)i - 2bj + bk] \cdot [-8i + 4j + 12k] &= 0 \\
 -8(a - 2) - 8b + 12b &= 0 \Rightarrow -2a + b + 4 = 0 \quad \dots (v)
 \end{aligned}$$

But the point $(1, -1, 2)$ lies on the surface (i), so

$$a + 2b - (a + 2) = 0 \Rightarrow 2b - 2 = 0 \Rightarrow b = 1$$

Putting the value of b in (v), we get

$$-2a + 1 + 4 = 0 \Rightarrow a = \frac{5}{2}$$

Hence, $a = \frac{5}{2}$, $b = 1$.

Example 10. Prove that $\vec{A} = (x^2 - yz)i + (y^2 - zx)j + (z^2 - xy)k$ is irrotational and find the scalar potential f such that $\vec{A} = \nabla f$.

$$\text{Sol. } \nabla \times \vec{A} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - yz & y^2 - zx & z^2 - xy \end{vmatrix} = (-x + x)i - (-y + y)j + (-z + z)k = 0$$

Hence, $\vec{A}$ is irrotational.

$$\text{Now } \vec{A} = \nabla f = i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z} = (x^2 - yz)i + (y^2 - zx)j + (z^2 - xy)k$$

Comparing on both sides, we get

$$\frac{\partial f}{\partial x} = (x^2 - yz), \quad \frac{\partial f}{\partial y} = (y^2 - zx) \text{ and } \frac{\partial f}{\partial z} = (z^2 - xy)$$

$$\begin{aligned} \therefore df &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = (x^2 - yz)dx + (y^2 - zx)dy + (z^2 - xy)dz \\ &= (x^2 dx + y^2 dy + z^2 dz) - (yzdx + zxdy + xydz) \end{aligned}$$

$$\text{or } df = \frac{1}{3} d(x^3 + y^3 + z^3) - d(xyz)$$

On integrating, we get

$$f = \frac{1}{3} (x^3 + y^3 + z^3) - xyz + c.$$

EXERCISE 5.3

1. Find $\text{div } \vec{A}$, when $\vec{A} = x^2zi - 2y^3z^2j + xy^2zk$. **[Ans. $2xz - 6y^2z^2 + xy^2$]**

2. If $\vec{V} = \frac{xi + yj + zk}{\sqrt{x^2 + y^2 + z^2}}$, find the value of $\text{div } \vec{V}$. (U.P.T.U., 2000)

$$\left[\text{Ans. } 2 \text{ division } \sqrt{(x^2 + y^2 + z^2)} \right]$$

3. Find the directional derivative of the divergence of $f(x, y, z) = xyi + xy^2j + z^2k$ at the point $(2, 1, 2)$ in the direction of the outer normal to the sphere, $x^2 + y^2 + z^2 = 9$. **[Ans. $\frac{13}{3}$]**

4. Show that the vector field $\vec{f} = \vec{r} / |\vec{r}|^3$ is irrotational as well as solenoidal. Find the scalar potential. (U.P.T.U., 2001, 2005) **[Ans. $-\frac{1}{\sqrt{x^2 + y^2 + z^2}}$]**

5. If r is the distance of a point (x, y, z) from the origin, prove that $\text{curl} \left(k \times \text{grad} \frac{1}{r} \right) + \text{grad} \left(k \cdot \text{grad} \frac{1}{r} \right) = 0$, where k is the unit vector in the direction OZ . (U.P.T.U., 2000)
6. Prove that $\vec{A} = (6xy + z^3)i + (3x^2 - z)j + (3xz^2 - y)k$ is irrotational. Find a scalar function $f(x, y, z)$ such that $\vec{A} = \nabla f$. [Ans. $f = 3x^2y + xz^3 - zy + c_2$]
7. Find the curl of $yz i + 3xz j + zk$ at $(2, 3, 4)$. [Ans. $-6i + 3j + 8k$]
8. If $f = x^2yz$, $g = xy - 3z^2$, calculate $\nabla \cdot (\nabla f \times \nabla g)$. [Ans. zero]
9. Determine the constants a and b such that curl of $(2xy + 3yz)i + (x^2 + axz - 4z^2)j + (3xy + 2byz)k = 0$. [Ans. $a = 3, b = -4$]
10. Find the value of constant b such that $\vec{A} = (bxy - z^3)i + (b - 2)x^2j + (1 - b)xz^2k$ has its curl identically equal to zero. [Ans. $b = 4$]
11. Prove that $\vec{a} \cdot \nabla \left(\frac{1}{r} \right) = \frac{\vec{a} \cdot \vec{r}}{r^3}$.
12. Prove that $\nabla (\vec{a} \cdot \vec{u}) = (\vec{a} \cdot \nabla)\vec{u} + \vec{a} \times \text{curl } \vec{u}$ is a constant vector.
13. Prove that $\nabla^2 \left(\frac{x}{r^3} \right) = 0$.
14. Prove that $\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$.
15. If $u = x^2 + y^2 + z^2$ and $\vec{v} = x\hat{i} + y\hat{j} + z\hat{k}$, show that $\text{div} (\vec{u}\vec{v}) = 5u$
16. Prove the curl $\left(\frac{\vec{a} \times \vec{r}}{r^3} \right) = -\frac{\vec{a}}{r^3} + \frac{3\vec{r}(\vec{a} \cdot \vec{r})}{r^5}$.
17. Find the curl of $\vec{v} = e^{xyz} (\hat{i} + \hat{j} + \hat{k})$ at the point $(1, 2, 3)$. [Ans. $e^6(i - 21j + 3k)$]
18. Prove that $\nabla \times \nabla f = 0$ for any $f(x, y, z)$.
19. Find curl of $\vec{A} = x^2yi - 2xzj + 2yzk$ at the point $(1, 0, 2)$. [Ans. $4\hat{j}$]
20. Determine curl of $xyz^2i + yzx^2j + zxy^2k$ at the point $(1, 2, 3)$.
21. Find $f(r)$ such that $f(r) \vec{r}$ is solenoidal. [Ans. $xy(2z - x)\hat{i} + yz(2x - y)\hat{j} + zx(2y - z)\hat{k}; 10\hat{i} + 3\hat{k}$
[Ans. $\frac{c}{r^3}$]
22. Find a, b, c when $\vec{f} = (x + 2y + az)i + (bx - 3y - z)j + (4x + cy + 2z)k$ is irrotational. [Ans. $a = 4, b = 2, c = -1$]
23. Prove that $(y^2 - z^2 + 3yz - 2x)i + (3xz + 2xy)i + (3xy - 2xz + 2z)k$ are both solenoidal and irrotational. [U.P.T.U., 2008]